Size selection (Purification) (1X ratio)

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Abstract

Size selection (Purification) with 1X ratio to keep more DNA than original version

Procedure

Step 1

Calculate the amount of magnetic beads you need and **DISTRIBUTE in a new 1.5 mL eppendorf tube (**Mix the bottle **VIGOROUSLY** to make sure it's homogeneous**).**

For example, if you have 5 samples to clean up, distribute 49.5 µl in a new 1.5 mL eppendorf tube.

Sample	Magnetic beads need
1	25 µl
2	50 µl
3	75 µl
4	100 µl
5	125 µl
6	150 µl
7	175 µl
8	200 µl

> **Note**
>
> The ratio is **25 µl of sample** use **25 µl of magnetic beads** to clean up (1 X to keep moderate to long DNA fragments).

> **Note**
>
> **Magnetic bead**: BeaverBeads™ DNA Select Isolation

Step 3 *(5 min)*

Mix sample and beads by **gently flicking** the tube for 5mins **5 min**

> **Note**
>
> Be gentle, mix too aggressive will make DNA fragmented

Step 4

Transfer the tube to the 8-strip magnetic rack.

After all the magnetic beads are arrested, remove the supernatant.

Step 5

Add **200 µL** 80% ethanol, flip the magnetic rack around to clean the pellet.

Wait until all the magnetic beads are arrested, remove the supernatant.

> **Note**
>
> **FRESHLY** make 80% ethanol

Step 6

Repeat the **step 5**.

Step 7

Flash spin the tube, and put on the magnetic rack.

Remove **ANY trace of liquid residuals** using 10 µl pipette.

> **Note**
>
> Caution: DO NOT let the beads crack

Step 8

Add **15 µL** elution buffer.

Mix gently by flicking the tube and make sure all magnetic beads are dissolved in buffer, then flash spin down the sample.

Step 10

Transfer the tube to the magnetic rack.

After all the magnetic beads are arrested, collect the **15 µL** supernatant to a 200 µl PCR tube or a 8-strip PCR tube.

Step 11

The clean-up sample is now ready for 2’ PCR.

Step 9 *(10 min)*

Incubate the tube in PCR thermal cycler using “**37_incubation**” program for 10min **10 min** .

Step 2

In each **25 µL** sample, add **25 µL** magnetic beads in.

> **Note**
>
> **Pipetting thoroughly every time** before adding to make sure all magnetic beads mixed well.